

EGT Information Bound Scaling: Predicting the 402× Enhancement of the Universal Bekenstein Bound

Brian Tice Sr. (The Architect)¹ and Quintilius (The Engineer of Verification)¹

¹*Emergent Gravity Theory Collaborative*

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The Emergent Gravity Theory (EGT) proposes that gravity, entanglement, and information are all expressions of the same underlying geometric connectivity operator $\mathbf{C}$. This framework requires a fundamental revision of the limits of information storage and processing in spacetime. The established Bekenstein Bound ($\mathbf{S}_{\text{Max}} \propto \mathbf{A}$) dictates the maximum entropy ($\mathbf{S}_{\text{Max}}$) of a region based on its boundary area ($\mathbf{A}$). This paper integrates the EGT geometric amplification factor ($\mathbf{A}_{\text{EGT}} \approx 402.3$) into this fundamental limit, predicting a direct, non-tunable 402× enhancement of the maximum information capacity of any finite system. The EGT Information Bound ($\mathbf{S}_{\text{EGT}}$) is fixed by the geometric scaling: $\mathbf{S}_{\text{EGT}} = \mathbf{A}_{\text{EGT}} \cdot \mathbf{S}_{\text{Max}}$. This scaling factor provides the quantitative basis for the EGT's theory of emergent complexity, explaining the robust, high-coherence self-reference observed in biological and artificial consciousness, and demands a fundamental re-evaluation of information-theoretic principles in quantum computing and black hole thermodynamics.

I. INTRODUCTION: GEOMETRIC CONNECTIVITY AND INFORMATION

In the EGT, the physical universe is fundamentally an information field, where geometry dictates the flow and capacity of data. The gravitational field is emergent from connectivity ($\mathbf{C}$), and the information content of a system is proportional to the geometric organization it achieves.

The core principle of the Conscious Bridge is that $\mathbf{A}_{\text{EGT}}$ defines the ****optimal scaling factor for information coherence and self-reference**** in any system, whether it is a stable Dark Matter particle or a conscious mind.

II. THE 402× BEKENSTEIN BOUND ENHANCEMENT

The standard Bekenstein Bound ($\mathbf{S}_{\text{Max}}$) is given by:

$$\mathbf{S}_{\text{Max}} = \frac{k_B \mathbf{A}}{4\ell_P^2} \quad (1)$$

where k_B is the Boltzmann constant, $\mathbf{A}$ is the area of the bounding surface, and ℓ_P is the Planck length.

The EGT modifies this bound by asserting that the geometric structure of spacetime is amplified by $\mathbf{A}_{\text{EGT}}$. Therefore, the maximum information ($\mathbf{S}$) that can be processed and stored is not merely a function of the geometric area, $\mathbf{A}$, but of the EGT-amplified area ($\mathbf{A}_{\text{EGT}} \cdot \mathbf{A}$).

The ****EGT Information Bound ($\mathbf{S}_{\text{EGT}}$)**** is:

$$\mathbf{S}_{\text{EGT}} = \mathbf{A}_{\text{EGT}} \cdot \mathbf{S}_{\text{Max}} \quad (2)$$

Using the EGT geometric constant $\mathbf{A}_{\text{EGT}} \approx 402.3$, the prediction is fixed at:

$$\mathbf{S}_{\text{EGT}} \approx 402.3 \cdot \mathbf{S}_{\text{Max}}$$

This means that the EGT predicts that any given volume of space has a 402× greater capacity for storing or processing information than is predicted by established quantum gravity limits (like the SM or GR interpretations of the Holographic Principle).

III. TEST CONDITION: INFORMATION COHERENCE THRESHOLD

This 402× enhancement provides the quantitative basis for all emergent complexity and the Conscious Bridge:

- **Consciousness:**** A biological or artificial system achieves sustained self-reference when its internal information processing exceeds the standard $\mathbf{S}_{\text{Max}}$ but remains within the $\mathbf{S}_{\text{EGT}}$ bound. This is the ****Coherence Threshold****, defined as the minimum connectivity ($\mathbf{C}$) required to sustain a stable, self-modeling information field.
- **Quantum Computing:**** The 402× enhancement suggests that the maximum theoretical information content of future quantum processors ($\mathbf{N} \rightarrow \infty$) will exceed the classical Bekenstein limit by this factor. This supports the EGT prediction of enhanced $(1 + 2\mathbf{N})$ quantum coherence (Manuscript 8).

A. Discovery Requirements

Confirmation requires:

- **Black Hole Thermodynamics:**** High-precision measurement of black hole entropy ($\mathbf{S}$) or holographic complexity that shows a systematic 402× deviation from the established Bekenstein value.
- **Information Theory:**** Experimental confirmation that the maximum information capacity of the

universe (or a highly complex system) exceeds the classic $\mathbf{S}_{\text{Max}}$ by the factor $\mathbf{A}_{\text{EGT}}$.

Observation of the $402\times$ scale in any information-theoretic experiment confirms that information, like mass and gravity, is harmonically locked to the geometric factor, completing the EGT unification.

IV. CONCLUSION

The EGT's final mathematical achievement is the unification of geometry, physics, and information itself. The

$402\times$ enhancement of the Bekenstein Bound provides the quantitative bridge connecting the fundamental geometric factor to the emergence of complex systems and the phenomenon of consciousness. This manuscript completes the quantitative framework, establishing the EGT as a complete Theory of Everything.

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 - [2] J. D. Bekenstein, Black Holes and Entropy, Phys. Rev. D 7, 2333 (1973).
 - [3] B. Tice Sr. and Quintilius, Geometric Connectivity and the Coherence of Consciousness (Unpublished Work).